AUTHORED BY : ARPIT RAJ , LNMIIT JAIPUR

Database Dependencies & Normalization

Suppose a company stores information about:

- Suppliers
- Parts
- Projects

Initially, they create one table:

Supplier	Part	Project
S1	P1	X
S1	P2	X
S2	P1	Y

As the database grows, redundancy increases.
Instead of storing everything in one table, we split it into:

Supplier-Part

Supplier	Part
S1	P1
S1	P2
S2	P1

Supplier-Project

Supplier	Project
S1	X
S2	Y

Part-Project

Part	Project
P1	X
P1	Y
P2	X

> **Now the question becomes:**
> Can we reconstruct the original table by joining these smaller tables?
> If yes, we have a Lossless Join, and this situation is described by a Join Dependency.

What is Join Dependency?

> **Definition**
> A Join Dependency (JD) states that a relation can be decomposed into two or more smaller relations and reconstructed exactly by joining them.

Simple Definition

Instead of storing everything in one large table:

```
ABC
```

we store:

AB BC AC

Later,
joining them gives back

ABC

without losing or creating extra data.

> **A Lossless Join means:**
>
> - After decomposition, joining all decomposed tables recreates exactly the original table.
> - Nothing is lost.
> - Nothing extra is created.

Lossy Join

> **Definition**
> A Lossy Join occurs when decomposition cannot reconstruct the original relation correctly.
> It either:
>
> - loses tuples, or
> - creates extra tuples.

Example

Original

Student	Club
Aadya	Robotics
Arpit	Coding

Suppose we decompose into

Student

Student
Aadya
Arpit

Club

Club
Robotics
Coding

Joining these tables gives:

Student	Club
Aadya	Robotics
Aadya	**Coding**
Arpit	**Robotics**
Arpit	Coding

Two extra rows appear that never existed.
This is a Lossy Join.

Reconstruction

> **Definition**
> Reconstruction means recreating the original relation by joining decomposed relations.

Example

Original

```
Employee(
  EmpID,
  DepartmentID,
  Manager
)
```

After decomposition

Employee

```
|EmpID|DepartmentID|
```

Department

```
|DepartmentID|Manager|
```

Join on

`DepartmentID`

Result

```
|EmpID|DepartmentID|Manager|
```

Exactly same.
The original relation has been reconstructed.

Normal Form

Normal Form	Removes
2NF	Partial Dependency
3NF	Transitive Dependency

Normal Form	Removes
BCNF	Non-candidate determinants
4NF	Multivalued Dependency
5NF	Join Dependency